Foresight: Cache-Friendly Skiplists for In-Memory Indexes
Supplementary Material

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Appendix A: Additional Cache Event Measurements

This appendix contains additional details and measurements of cache events that were excluded from the main paper due to space constraints. The hardware cache events we used to monitor cache references and misses are listed below:

- L1 references = L1-dcache-loads + L1-dcache-stores.
- L1 misses = L1-dcache-load-misses.
- L3 references (L2 misses) = LLC-loads + LLC-stores.
- L3 misses = LLC-load-misses + LLC-store-misses.

The additional measurements included in this appendix are:

1. A breakdown of cache misses per operation across cache levels and workloads with 2^{25} skiplist elements and a varying number of participating threads (Fig. 1).
2. The amount of references to the L1 data cache per skiplist operation in all experiments (Figures 2, 3).
3. The percentage of misses in all cache levels out of the total references to the L1 data cache (Fig. 4, 5).

Figure 1 clearly demonstrates the superiority of *Foresight* in terms of reduced cache misses when the skiplist contains a large number of elements.

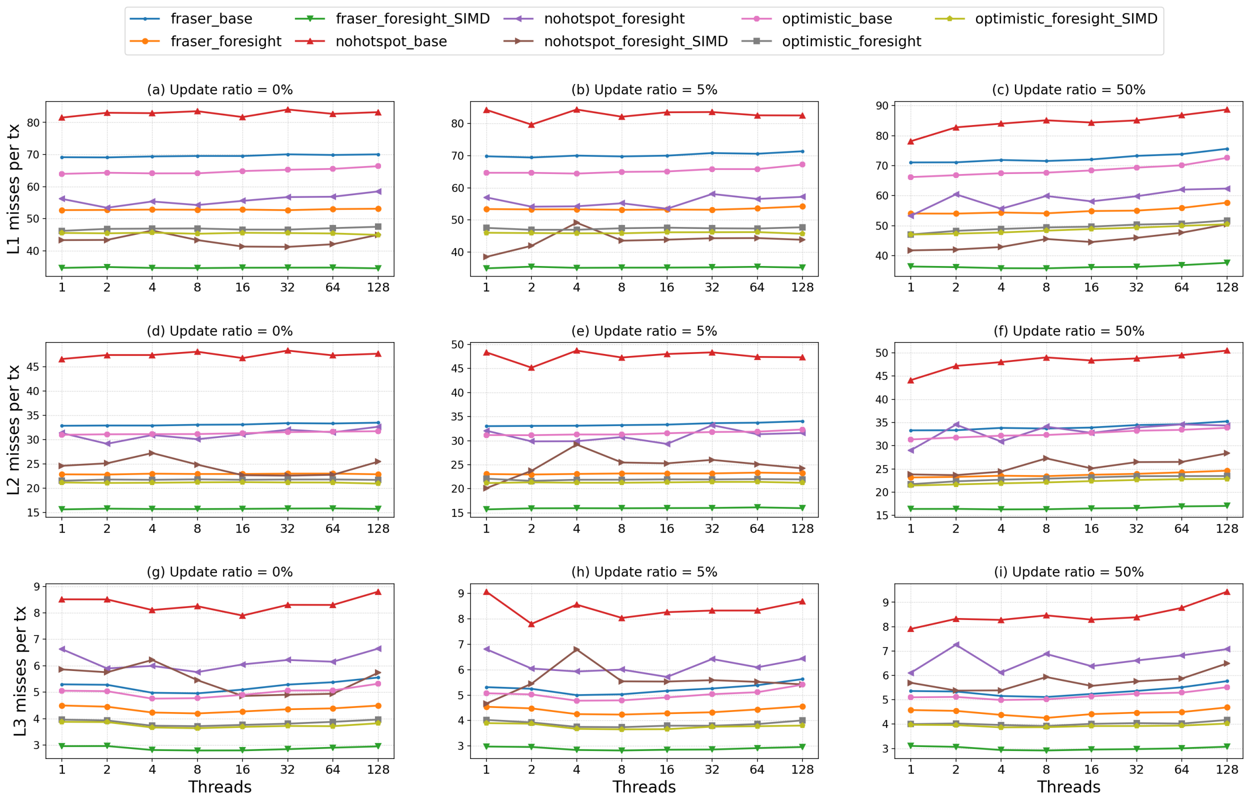

Figure 1: Cache misses per skiplist operation with data structure size of 2^{25} elements, varying number of participating threads. The columns correspond to the different microbenchmarks and the rows correspond to the different cache levels.

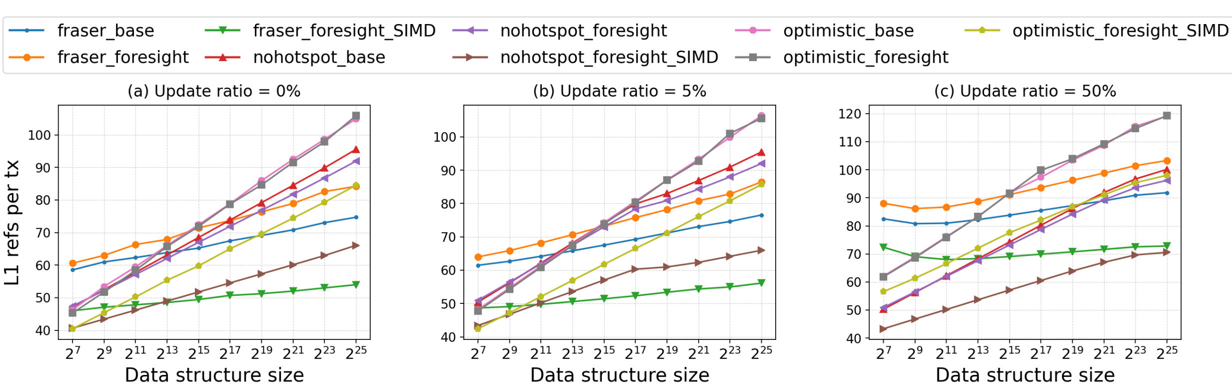

Figure 2: References to the L1 data cache per skiplist operation. 128 threads, varying data structure sizes.

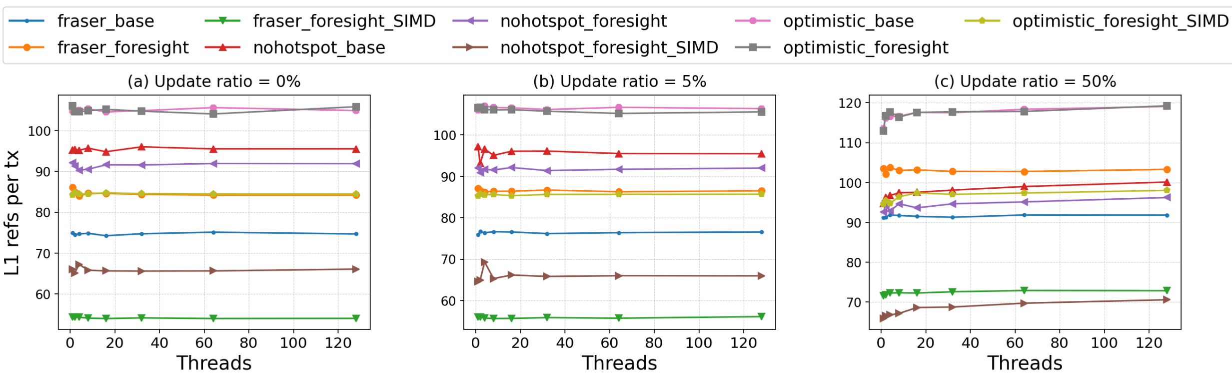

Figure 3: References to the L1 data cache per skiplist operation. 2^{25} skiplist elements, varying number of threads.

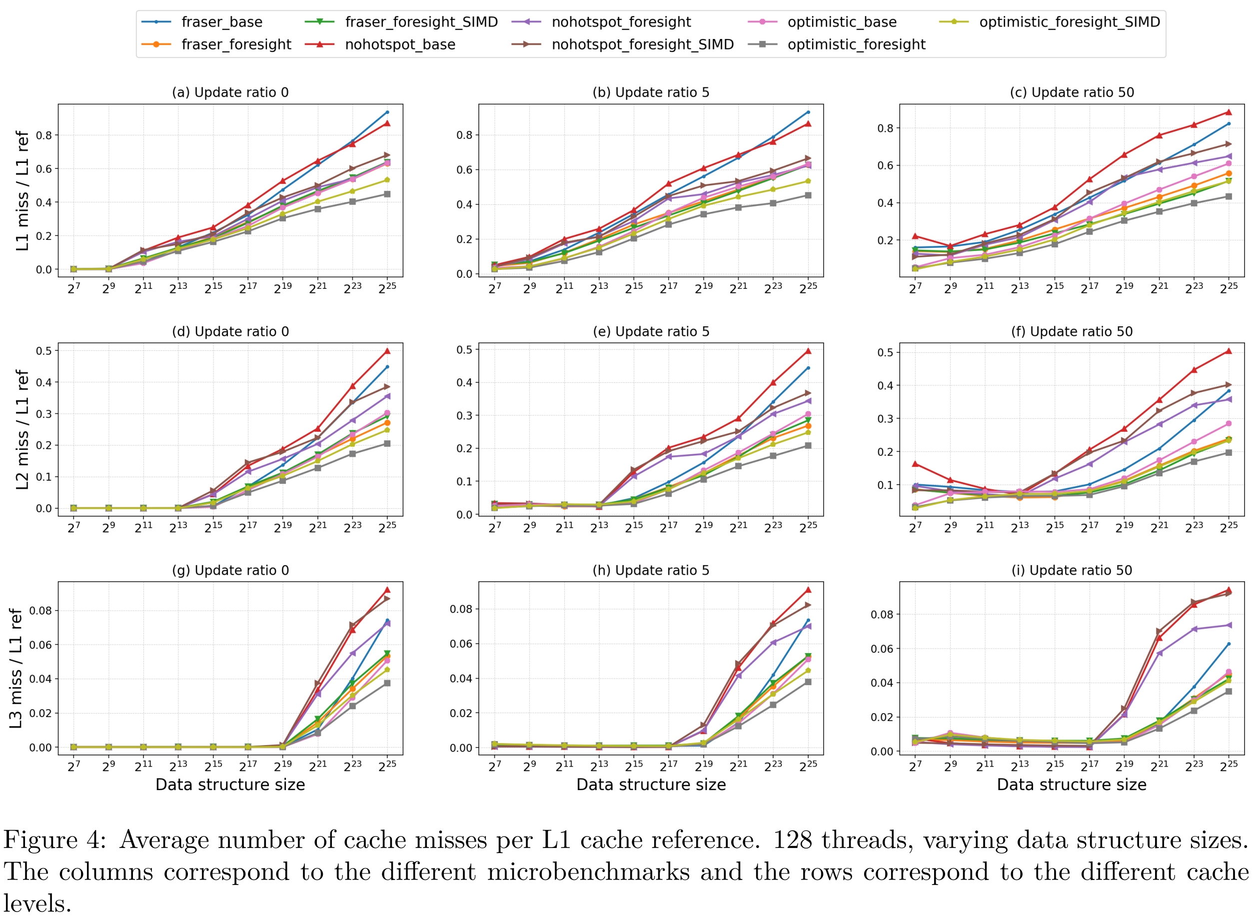

Figure 4: Average number of cache misses per L1 cache reference. 128 threads, varying data structure sizes. The columns correspond to the different microbenchmarks and the rows correspond to the different cache levels.

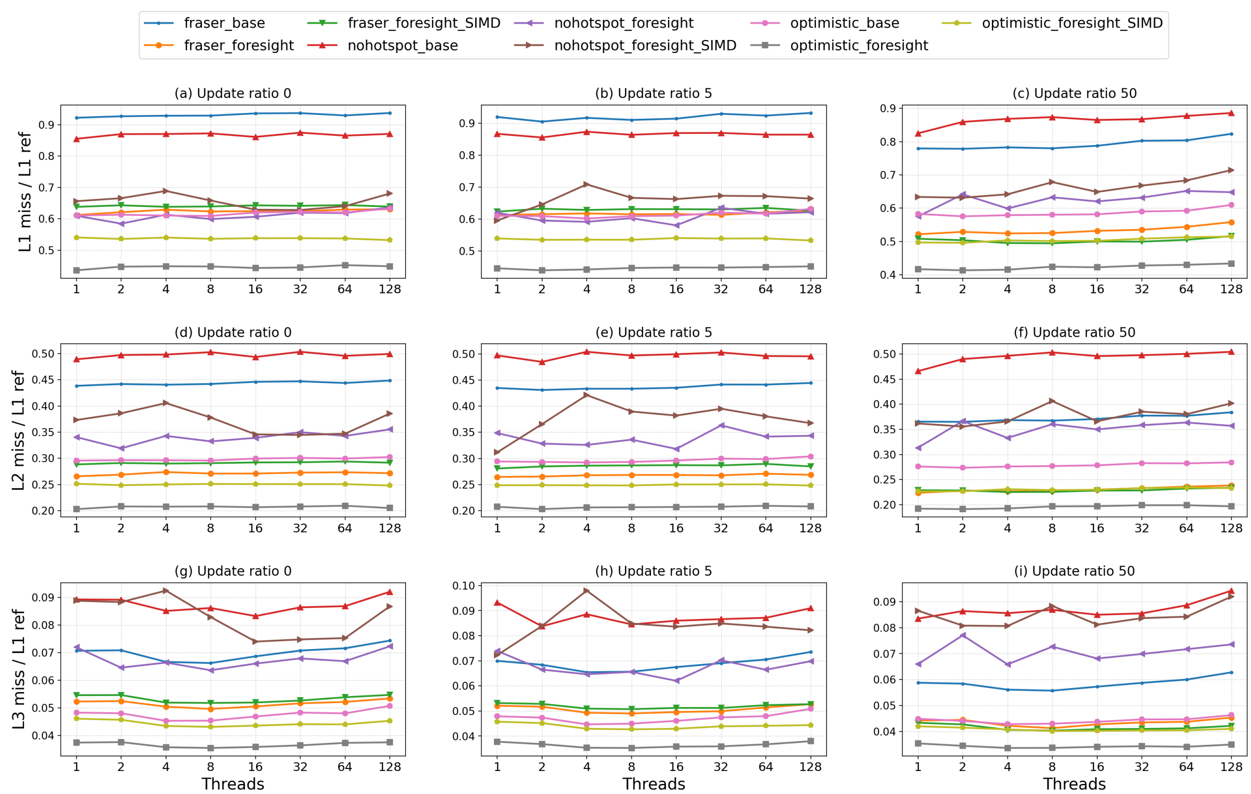

Figure 5: Average number of cache misses per L1 cache reference. 2^{25} skiplist elements, varying number of threads. The columns correspond to the different microbenchmarks and the rows correspond to the different cache levels.